# Topic 37 — Greenhouse Gases

**Covers syllabus (click to expand)**

Carbon Dioxide ($CO_2$) (कार्बन डाइऑक्साइड) | Methane ($CH_4$) | Nitrous Oxide ($N_2O$) | Water Vapour | Ozone | Chlorofluorocarbons (CFCs) (सीएफसी) | Hydrofluorocarbons (HFCs) (एचएफसी)

**Sources baked in:** IPCC AR6; NCERT; Kyoto basket; Montreal/Kigali; UPPCS PYQs **Weight:** ★★★ High — Argon≠GHG; paddy→$CH_4$; laughing gas=$N_2O$; CFC dual role; biogas $CH_4$+$CO_2$ **Last verified:** August 2026

## Current Affairs (this topic)

- WMO GHG Bulletin and methane CA refresh this twin of Climate Change (जलवायु परिवर्तन).
- GWP order traps ($CO_2$ / $CH_4$ / $N_2O$ / fluorinated) stay core.

## 37.1 Carbon Dioxide ($CO_2$)

**Identity:** Dominant by volume | GWP = 1 | Fossil fuels / cement / deforestation

- **$CO_2$** is the main anthropogenic GHG by total forcing (~**76%** IPCC-style). Many Indian papers also use WMO-style relative shares: **~64% $CO_2$, ~19% $CH_4$, ~6% $N_2O$, ~11% CFCs + others**.
- **GWP = 1** — the reference gas for converting other gases to **$CO_2$e**.
- Pre-industrial level ~**280 ppm (~0.03%)**; now **415–420+ ppm**.
- Main human sources: **fossil fuel combustion**, **cement**, **deforestation / land-use change**.
- Persists in the climate system for a very long time compared with methane.
- India ranks among the top absolute emitters (~**3rd**; **China** #1) but has **low per capita** emissions.
- **Carbon fertilization** increases plant growth from higher $CO_2$ — distinct from warming and ocean acidification (महासागरीय अम्लीकरण).
- $CO_2$ forms weak carbonic acid in rain (natural pH ~5.6) but is **not** the main driver of damaging acid rain (अम्ल वर्षा) ($SO_2$/NOx dominate).
- $CO_2$ is both **beneficial** (photosynthesis) and **harmful** (enhanced greenhouse warming).

**PYQ — UPPCS Prelims 2018, Q132**

Largest $CO_2$ emitter in 2015:

Options: A. China B. USA C. India D. France

**Show answer**

**Ans: A**

> **Logic:** Highest GWP ≠ most total warming. **$CO_2$ wins on volume.**

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## 37.2 Methane ($CH_4$)

**Identity:** Marsh gas | GWP ~28–36 | Paddies / cattle / wetlands | Hydrates

- **Methane** is the second major anthropogenic GHG by forcing (~**16%** IPCC-style; ~**19%** in WMO-style relative-share stems).
- Also called **marsh gas**.
- **GWP ≈ 28–36** (100-year); atmospheric lifetime ~**12 years**. Short-horizon GWP is much higher than $CO_2$.
- Flooded **rice paddies** create anaerobic soils that emit $CH_4$. With nitrogen fertiliser, **rice** is also a major **$N_2O$** source.
- **Livestock** enteric fermentation (cattle/buffalo) is a major Indian source.
- Other sources: **landfills**, coal mines, natural **wetlands** (~**75–76%** of natural $CH_4$), and **termites**.
- **Methane hydrates** under Arctic tundra (टुंड्रा) and seafloor can release $CH_4$ if warmed; $CH_4$ oxidises to $CO_2$ in roughly a decade or two.
- **CNG** is mainly **methane** (~90%).
- **Biogas** is mainly **methane + carbon dioxide**.

**PYQ — UPPCS Prelims 2019, Q62**

Gas emitted in paddy fields that increases Earth's temperature:

Options: A. Nitrogen B. Carbon dioxide C. Carbon monoxide D. Methane

**Show answer**

**Ans: D**

**PYQ — UPPCS Prelims 2018, Q63**

Methane is emitted from:

Options: A. Paddy fields only B. Termitary only C. Both D. None

**Show answer**

**Ans: C**

**PYQ — UPPCS Prelims 2022, Q113**

CNG is mainly composed of—

Options: A. Methane B. Propane C. Ethane D. Butane

**Show answer**

**Ans: A**

**PYQ — UPPCS Prelims 2025, Q28**

Major constituents of biogas:

1. Methane 2. Butane 3. Carbon dioxide 4. Carbon monoxide

Options: A. 1 and 3 B. 3 and 4 C. 2 and 3 D. 1 and 2

**Show answer**

**Ans: A** — **$CH_4$ + $CO_2$**.

> **Logic:** Paddy temperature gas = **methane**, not CO or $N_2$.

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## 37.3 Nitrous Oxide ($N_2O$)

**Identity:** Laughing gas | GWP ~265 | Fertilizer soils

- **$N_2O$** is a long-lived GHG with **GWP ≈ 265** (100-year) and lifetime ~**100+ years**.
- It contributes about **~6%** of anthropogenic GHG forcing.
- Main human source: **nitrogen fertilisers** in soils (nitrification/denitrification).
- Also from combustion, biomass burning, and some industry.
- Common name: **laughing gas** (medical anaesthetic use).
- Do not confuse with **NO/$NO_2$** (air pollutants) or **ammonia ($NH_3$)**.
- In Kyoto climate basket; **not** a classic Montreal ODS.

**PYQ — UPPCS Prelims 2023, Q105**

Which is known as Laughing Gas?

Options: A. Ammonia B. Nitrous Oxide C. Carbon Monoxide D. Chlorofluorocarbon

**Show answer**

**Ans: B**

> **Logic:** Laughing gas = **$N_2O$ only**.

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## 37.4 Water Vapour

**Identity:** Most abundant natural GHG | Feedback, not primary driver

- **Water vapour** is the strongest contributor to the **natural** greenhouse effect by volume.

- It is mainly a **feedback**: warming increases evaporation → more vapour → more warming.
- Humans do not control water vapour the way they control $CO_2$/$CH_4$/$N_2O$ emission inventories.
- Concentration varies widely (about **0–4%**) with place and weather; most is in the troposphere (क्षोभमंडल).
- Condenses into clouds; linked to the hydrological cycle.

> **Logic:** "Largest anthropogenic GHG" is **not** water vapour — that label belongs to **$CO_2$** by forcing share.

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## 37.5 Ozone (as Greenhouse Gas)

**Identity:** Tropospheric $O_3$ = GHG + pollutant | Stratospheric $O_3$ = UV shield

- **Tropospheric (ground-level) ozone** is both an air pollutant and a greenhouse gas.
- It forms in photochemical smog from **NOx + VOCs + sunlight**.
- **Stratospheric ozone** is the UV shield — its ma role is protection, not climate forcing.
- "Good up, bad down" remains the safest fact.

**PYQ — UPPCS Prelims 2022, Q46**

Photochemical smog always contains—

Options: A. Phosphorus B. Ozone C. Methane D. Aluminium ion

**Show answer**

**Ans: B**

**PYQ — UPPCS Prelims 2020, Q139**

NOT a greenhouse gas found naturally (among options):

Options: A. Nitrogen oxide B. $CO_2$ C. Methane D. Ozone

**Show answer**

**Ans: A** — $CO_2$, $CH_4$, and $O_3$ are treated as natural GHGs here.

> **Logic:** Ozone can be a GHG **and** a pollutant — layer decides the story.

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## 37.6 Chlorofluorocarbons (CFCs)

**Identity:** Synthetic GHG + ODS | Very high GWP | Montreal phase-out

- **CFCs** are human-made compounds once used in refrigeration, aerosols, and foam blowing.

- They are **both** greenhouse gases (**very high GWP**) and **ozone-depleting substances**.
- Controlled and largely phased out under the **Montreal Protocol** (मॉन्ट्रियल प्रोटोकॉल).
- In the atmosphere they are long-lived; UV in the stratosphere (समतापमंडल) releases chlorine that destroys $O_3$.
- Still appear in "NOT a GHG?" questions as a **true GHG** distractor next to argon.

**PYQ — UPPCS Prelims 2022, Q102**

Which is NOT a greenhouse gas?

Options: A. Argon B. CFC C. Methane D. Carbon dioxide

**Show answer**

**Ans: A** — CFC **is** a GHG.

**PYQ — UPPCS Prelims 2025, Q45**

Ozone depletion (ओजोन क्षरण) has been linked to CFCs — **TRUE** (with UV-protection statement).

> **Logic:** CFC = climate warmer **and** ozone destroyer.

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## 37.7 Hydrofluorocarbons (HFCs)

**Identity:** CFC substitutes | GHG only | Kigali 2016

- **HFCs** replaced many CFCs/HCFCs in air-conditioning and refrigeration.
- They **do not deplete ozone** (no chlorine/bromine ODS role) but have **very high GWP**.
- Included in the **Kyoto** (क्योटो) climate basket; phased down under the **Kigali Amendment (2016)** (किगाली संशोधन) to the Montreal Protocol.
- Key fact: **HFC = hot but no hole**.

| Gas | ODS? | Climate role | Treaty home |
|---|---|---|---|
| CFC | Yes | High GWP GHG | Montreal |
| HFC | No | High GWP GHG | Kyoto + Kigali |
| HCFC | Yes (lower) | Transitional | Montreal phase-out |

> **Logic:** Replacing CFCs with HFCs fixed ozone risk but created a climate problem.

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